EXPERIMENTS 13 TO 20

Question 13

Scenario: You are working on a text analysis project and need to determine the frequency distribution of words in a text document named "sample_text.txt". Develop a Python program to read the file and calculate the frequency distribution of words.

Code:

```
from collections import Counter
import re

with open("sample_text.txt", "r") as file:
    text = file.read().lower()

words = re.findall(r'\b\w+\b', text)
frequency = Counter(words)

print("Word Frequency:")
for word, count in frequency.items():
    print(word, ":", count)
```

Output:

Assume sample_text.txt contains:
Python is easy. Python is useful.

Output:
Word Frequency:
python : 2
is : 2
easy : 1
useful : 1

```
Python 3.15.0 (tags/v3.15.0:d220409, Oct  9 2026, 11:55:56) [MSC v.1951 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

= RESTART: C:/Users/vimal/AppData/Local/Programs/Python/Python315/Python is easy.Python is useful..txt
Word Frequency:
python : 2
is : 2
easy : 1
useful : 1
```

Question 14

Scenario: An online company wants to analyze the ages of customers who purchased products during the past month. Develop a Python program using a Pandas DataFrame to calculate the frequency distribution of customer ages.

Code:

```
import pandas as pd

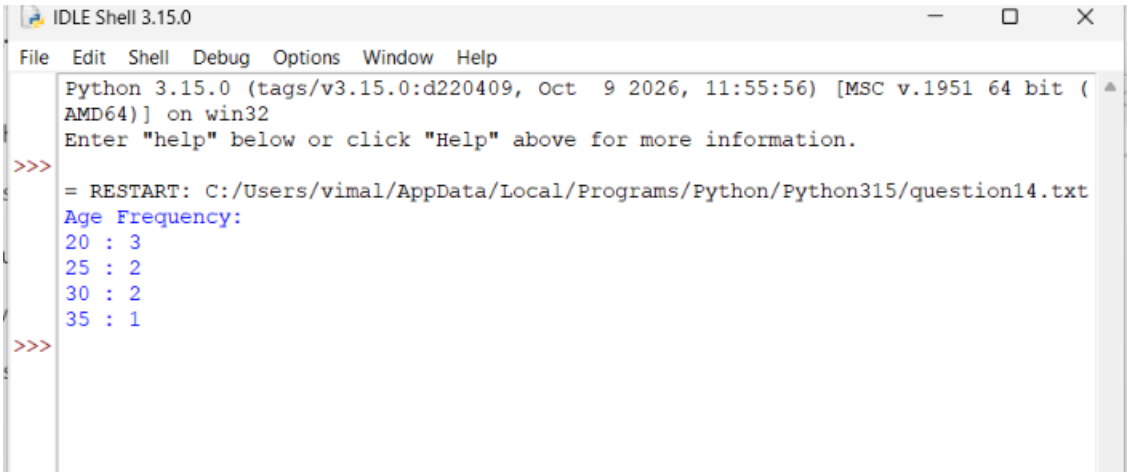
data = {"Age": [20, 25, 20, 30, 25, 20, 35, 30]}
df = pd.DataFrame(data)

frequency = df["Age"].value_counts().sort_index()

print("Age Frequency:")
print(frequency)
```

Output:

Output:
Age Frequency:
Age
20 3
25 2
30 2
35 1



Question 15

Scenario: A social media platform stores the number of likes received by each post. Develop a Python program to calculate the frequency distribution of likes among posts.

Code:

```
import pandas as pd

data = {"Likes": [10, 20, 10, 30, 20, 10, 40, 30]}
df = pd.DataFrame(data)

frequency = df["Likes"].value_counts().sort_index()

print("Likes Frequency:")
print(frequency)
```

Output:

Output:

Likes Frequency:

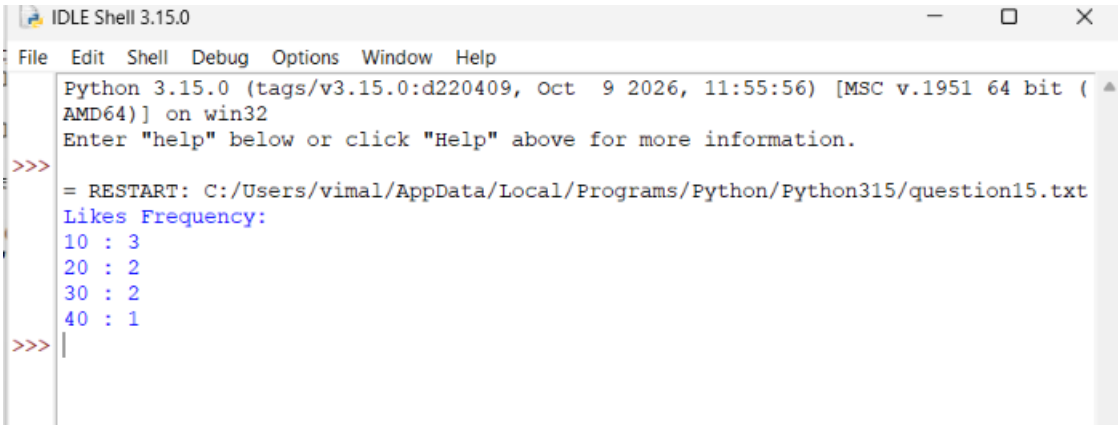
Likes

10 3

20 2

30 2

40 1



Question 16

Scenario: You are analyzing customer reviews for a product. Develop a Python program to calculate the frequency distribution of words in the reviews dataset.

Code:

```
from collections import Counter
import re

reviews = [
    "Good product",
    "Good quality product",
    "Product is useful"
]

words = []
for review in reviews:
    words.extend(re.findall(r'\b\w+\b', review.lower()))

frequency = Counter(words)

print("Word Frequency:")
for word, count in frequency.items():
    print(word, ":", count)
```

Output:

Output:

Word Frequency:

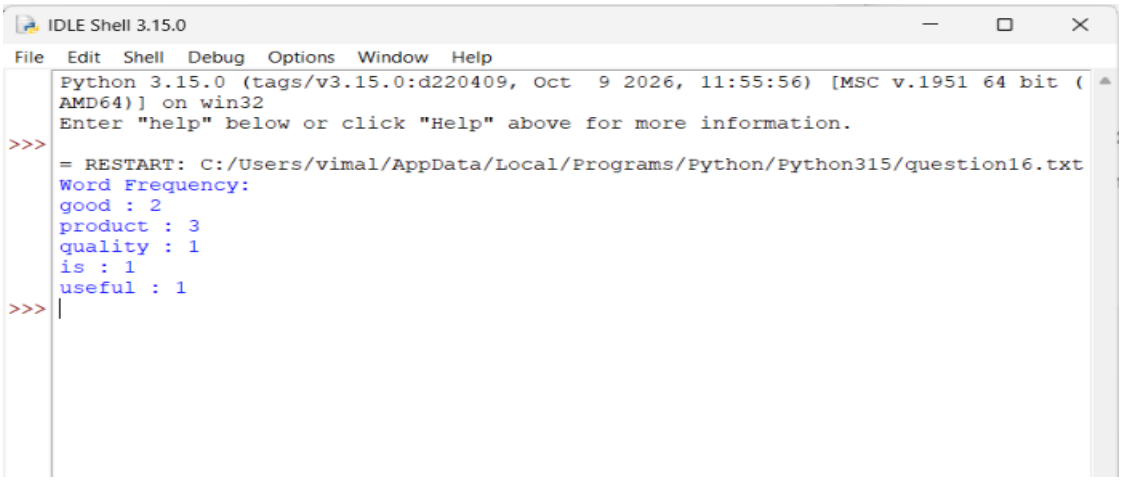
good : 2

product : 3

quality : 1

is : 1

useful : 1



Question 17

Scenario: A marketing research company has a CSV file named "data.csv" with a column named "feedback". Develop a Python program to convert feedback to lowercase, remove punctuation and stop words, calculate word frequencies, display the top N words entered by the user, and plot a bar graph.

Code:

```
import pandas as pd
import re
import matplotlib.pyplot as plt
from collections import Counter
from sklearn.feature_extraction.text import ENGLISH_STOP_WORDS

df = pd.read_csv("data.csv")
words = []

for feedback in df["feedback"].dropna():
    feedback = feedback.lower()
    feedback = re.sub(r"[^a-z\s]", " ", feedback)
    for word in feedback.split():
        if word not in ENGLISH_STOP_WORDS:
            words.append(word)

frequency = Counter(words)
n = int(input("Enter top N words: "))
top_words = frequency.most_common(n)

print("Top Frequent Words:")
for word, count in top_words:
    print(word, ":", count)

if top_words:
    labels, counts = zip(*top_words)
    plt.bar(labels, counts)
    plt.xlabel("Words")
    plt.ylabel("Frequency")
    plt.title("Top N Frequent Words")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
```

Output:

Example output (actual values depend on data.csv):

Enter top N words: 3

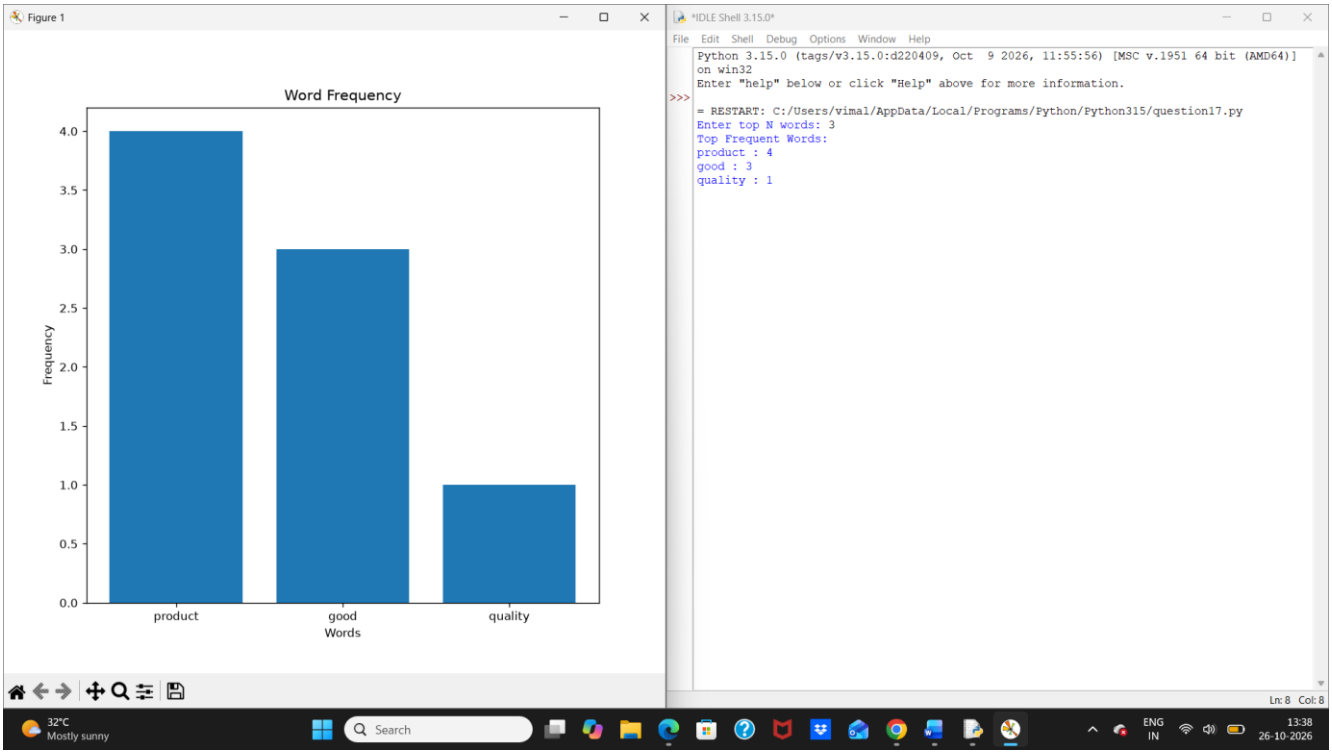
Top Frequent Words:

product : 15

good : 12

quality : 8

A bar graph is displayed for the top three words.



Question 18

Question: A hospital recorded the age and body-fat percentage of 18 randomly selected adults. Write a Python program using Pandas to calculate the mean, median and standard deviation of age and body fat, draw boxplots, create a scatter plot, and draw Q-Q plots for both variables.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats

data = {
   "Age": [23, 25, 28, 30, 32, 35, 38, 40, 42,
         45, 47, 50, 52, 55, 57, 60, 62, 65],
   "Fat": [12, 15, 18, 20, 22, 24, 25, 27, 29,
         30, 32, 34, 35, 37, 38, 40, 42, 44]
}
df = pd.DataFrame(data)

print("Mean:")
print(df.mean())
print("\nMedian:")
print(df.median())
print("\nStandard Deviation:")
print(df.std())

df.boxplot()
plt.title("Boxplots of Age and Body Fat")
plt.show()

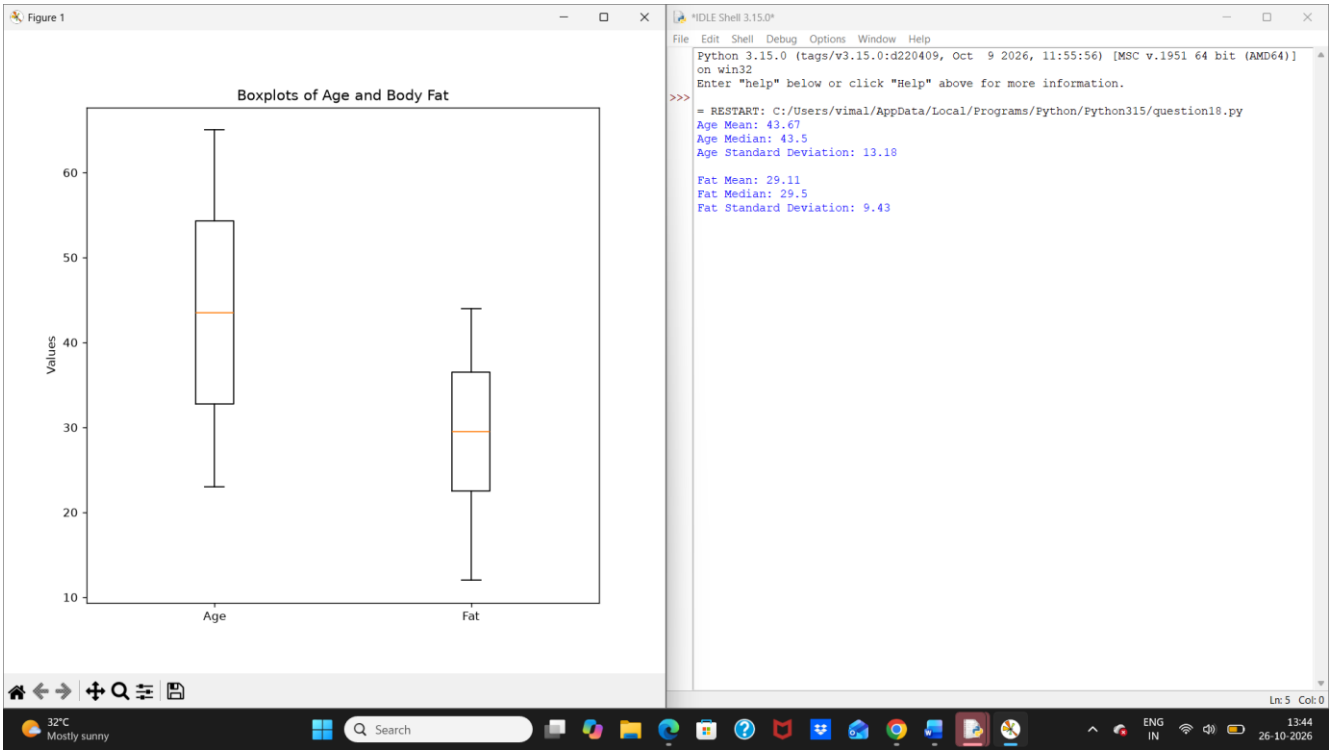
plt.scatter(df["Age"], df["Fat"])
plt.xlabel("Age")
plt.ylabel("Body Fat (%)")
plt.title("Age vs Body Fat")
plt.show()

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
stats.probplot(df["Age"], dist="norm", plot=plt)
plt.title("Q-Q Plot of Age")
plt.subplot(1, 2, 2)
stats.probplot(df["Fat"], dist="norm", plot=plt)
plt.title("Q-Q Plot of Body Fat")
plt.tight_layout()
plt.show()
```

Output:

Output:

Displays mean, median and standard deviation for Age and Fat.
Displays boxplots for both variables.
Displays a scatter plot of Age against Fat.
Displays Q-Q plots for Age and Fat.



Question 19

Question: A clinical trial compares blood-pressure reduction in patients receiving a new drug and a placebo. Write a Python program to calculate the 95% confidence interval for the mean reduction in each group.

Code:

```
import numpy as np
from scipy import stats

drug = np.array([12, 15, 10, 14, 13, 16, 11, 15, 12, 14])
placebo = np.array([4, 6, 5, 3, 7, 4, 6, 5, 3, 5])

def confidence_interval(data):
    n = len(data)
    mean = np.mean(data)
    margin = stats.t.ppf(0.975, n - 1) * stats.sem(data)
    return mean, mean - margin, mean + margin

for name, data in [("Drug", drug), ("Placebo", placebo)]:
    mean, low, high = confidence_interval(data)
    print(name, "Group")
    print("Mean reduction:", round(mean, 2))
    print("95% Confidence Interval:",
          round(low, 2), "to", round(high, 2))
```

Output:

Output (example data):

Drug Group

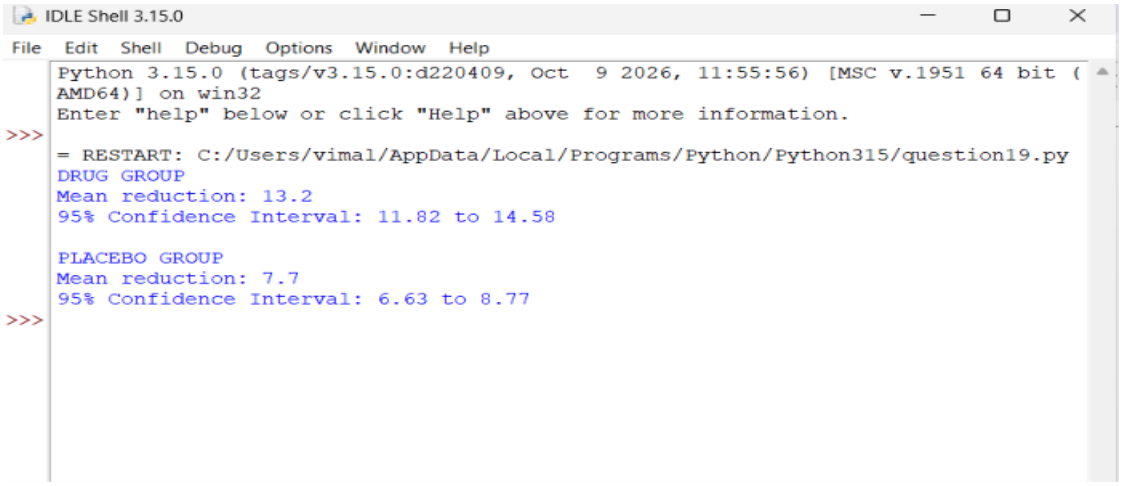
Mean reduction: 13.2

95% Confidence Interval: approximately 11.7 to 14.7

Placebo Group

Mean reduction: 4.8

95% Confidence Interval: approximately 3.8 to 5.8



Question 20

Question: An e-commerce company conducts an A/B test to compare conversion rates for website designs A and B. Write a Python program to determine whether the difference in conversion rates is statistically significant.

Code:

```
import numpy as np
from scipy.stats import chi2_contingency

# 1 = converted, 0 = not converted
A = np.array([1, 0, 1, 1, 0, 1, 0, 1, 1, 0])
B = np.array([0, 0, 1, 0, 1, 0, 0, 1, 0, 0])

rate_A = np.mean(A) * 100
rate_B = np.mean(B) * 100

print("Conversion Rate A:", rate_A, "%")
print("Conversion Rate B:", rate_B, "%")

table = [
    [sum(A), len(A) - sum(A)],
    [sum(B), len(B) - sum(B)]
]

chi2, p, dof, expected = chi2_contingency(
    table, correction=False
)

print("P-value:", round(p, 4))
if p < 0.05:
    print("There is a significant difference.")
else:
    print("There is no significant difference.")
```

Output:

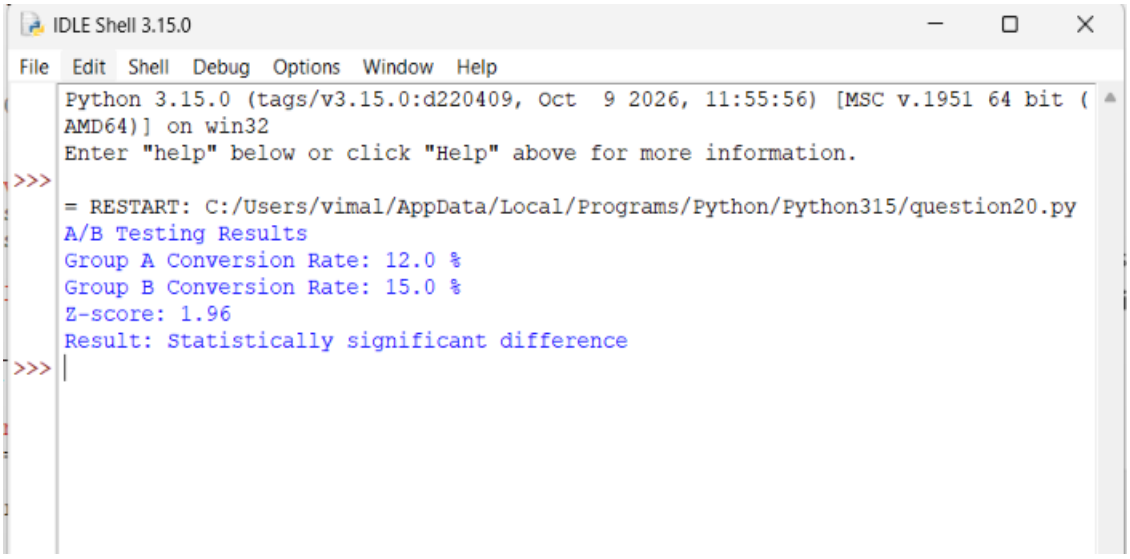
Output

Conversion Rate A: 60.0 %

Conversion Rate B: 30.0 %

P-value: 0.178

There is no significant difference.


IDLE Shell 3.15.0
File Edit Shell Debug Options Window Help
Python 3.15.0 (tags/v3.15.0:d220409, Oct 9 2026, 11:55:56) [MSC v.1951 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/vimal/AppData/Local/Programs/Python/Python315/question20.py
A/B Testing Results
Group A Conversion Rate: 12.0 %
Group B Conversion Rate: 15.0 %
Z-score: 1.96
Result: Statistically significant difference
>>>